

github username : orclitishra-15

$$E = 122433$$

$$\text{Coins} = 43$$

$$n = 10 + (122433 \% 90)$$

$$= 10 + 33 = 43$$

- My guess for 43 coins: using ~~something~~ binary search sort of thing where we divide the pile into 2 groups and chose the heavier one and repeat.

- Worst case for binary search (for 43 coins) $\rightarrow 5$

Step	Coins of scale	Worst case	Remaining
1	21-21	21 coins	1
2	10-10	10 coins	1
3	5-5	5 coin	0
4	2-2	2 coins	1
5	1-1	found	0

$\therefore$ Total worst case steps = 5

Keeping halving the coins on each scale.

- Now, up until now we've used 2 information given by weighing scale, i.e., right heavy or left heavy,

We can use one more information - balanced; by changing the base from 2 to 3.

- The idea: Divide the pile into three parts instead of two.
- If the left is heavier remove the rest two and repeat the same step on left, and vice versa.
- If both are equal then it must be in the third pile kept outside. So remove left and right and repeat the same process on the outside again.
- Base 3 works better than base 2 because it is using more of information and is dividing the pile even further.

Steps	on scale	Outside	worst case	
1	14-14	15	15	
2	5-5	5	5	
3	2-2	1	2	
4	1-1	0	found	

- What proves that this is the best & none other can do it better?

We can prove that base 3 would be the optimum solution by using "Information Theoretic Lower Bound" from information theory.

Proof: Each ^{weighing} coin will have either of 3 possible outcome.

$$1 \rightarrow 3$$

$$2 \rightarrow 3$$

$$3 \rightarrow 27$$

$$\Rightarrow w \rightarrow 3^w$$

$3^w \geq N$ (since each of the weighing ~~of~~ would have it's own outcome)

$\therefore$ for $N = 43$, w must be ≥ 4 , else by pigeon hole principle we can prove that it won't be correct.

what surprised me -

Changing the base can differ the outcome vastly.
And also the pigeon hole can be applied in weird places.

Did I use AI? For what?

Yes, to read the Information theory paper and give me the proof of why only base 3 is optimum, and I didn't wrote it.

The paper was on "Information Theoretic Lower bound"